

1. 基本概念

$$F^e = k^e \Delta^e$$

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8-1 单元刚度矩阵是指下列两组量值之间关系的变换矩阵。(A) (北京科技大学 2010、中国地震局工程力学研究所 2009、中南大学 2010)

A. 杆端位移和杆端力

B. 杆端位移和结点位移

C. 杆端位移和结点荷载

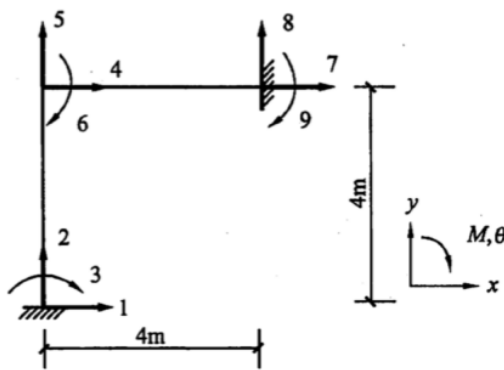
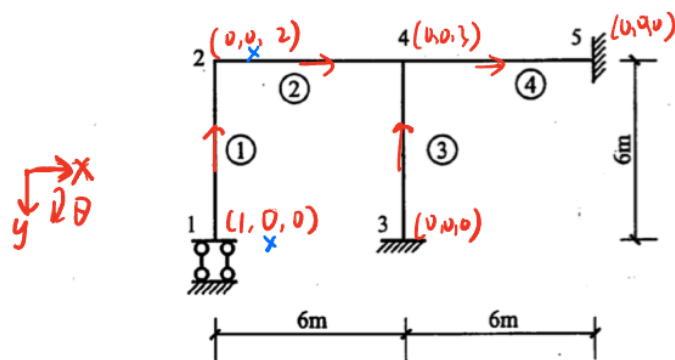
D. 结点位移和结点荷载

8-2 判断题。结构刚度矩阵反映了结构结点位移与荷载之间的关系。(X) (烟台大学 2013)

2. 刚度矩阵的计算

8-3 试用先处理法建立图 8-43 所示结构的结构刚度矩阵，忽略杆件的轴向变形 ($EI=常数$)。(湖南大学 2008)

8-4 求图 8-44 所示刚架结构刚度矩阵的任意两个主元素 (自由结点)。已知各杆 $I/A=1/10$ 。(西南交通大学 2010)



$$I/A = 1/10$$

$$k_1 = \begin{bmatrix} \frac{12EI}{l^3} & \frac{6EI}{l^2} & -\frac{12EI}{l^3} & \frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{4EI}{l} & -\frac{6EI}{l^2} & \frac{2EI}{l} \\ \frac{12EI}{l^3} & \frac{6EI}{l^2} & -\frac{12EI}{l^3} & \frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{2EI}{l} & -\frac{6EI}{l^2} & \frac{4EI}{l} \end{bmatrix}$$

$$k_2 = \begin{bmatrix} \frac{12EI}{l^3} & \frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{4EI}{l} \end{bmatrix}$$

$$k_3 = \begin{bmatrix} \frac{4EI}{l} & \frac{2EI}{l} \\ \frac{2EI}{l} & \frac{4EI}{l} \end{bmatrix}$$

$$k_4 = \begin{bmatrix} \frac{4EI}{l} & \frac{2EI}{l} \\ \frac{2EI}{l} & \frac{4EI}{l} \end{bmatrix}$$

$$k_5 = \begin{bmatrix} \frac{4EI}{l} & \frac{2EI}{l} \\ \frac{2EI}{l} & \frac{4EI}{l} \end{bmatrix}$$

$$K = \begin{bmatrix} \frac{12EI}{l^3} & \frac{6EI}{l^2} & 0 \\ \frac{6EI}{l^2} & \frac{8EI}{l} & \frac{2EI}{l} \\ 0 & \frac{2EI}{l} & \frac{12EI}{l^3} \end{bmatrix} = \begin{bmatrix} \frac{EI}{18} & \frac{EI}{6} & 0 \\ \frac{EI}{6} & \frac{4EI}{3} & \frac{EI}{3} \\ 0 & \frac{EI}{3} & \frac{2EI}{3} \end{bmatrix}$$



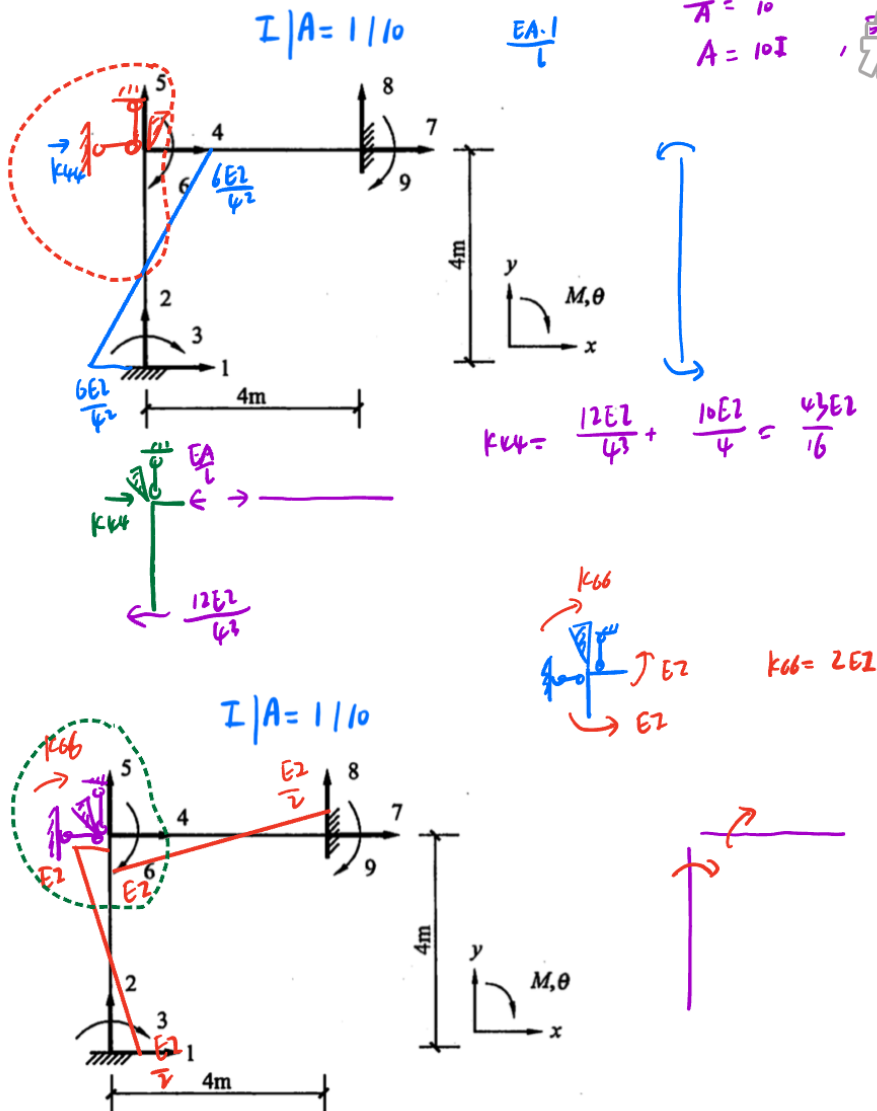
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8-4

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$$\frac{I}{A} = \frac{1}{10}$$

$$A = 10I$$

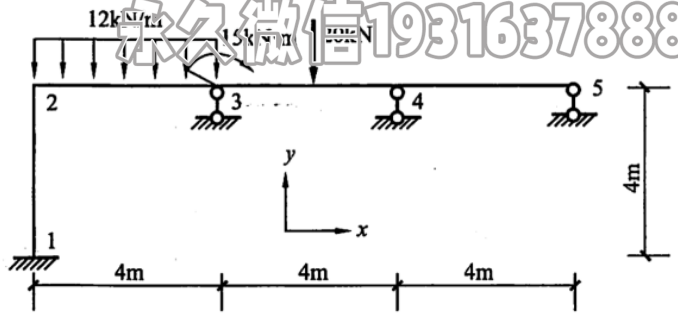
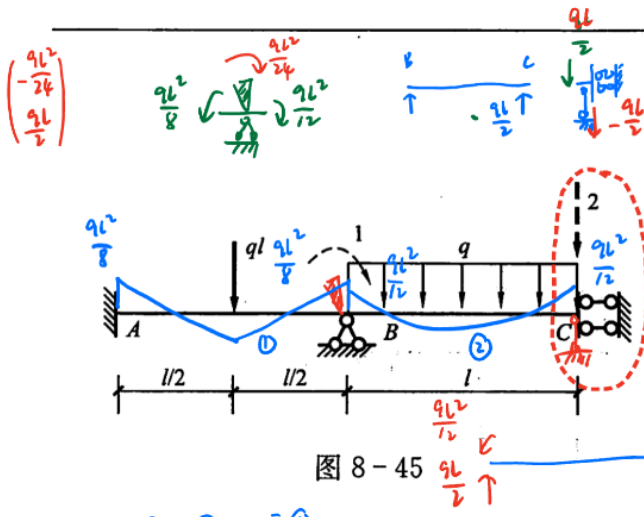


3. 等效结点荷载

8-5 图 8-45 所示结构按矩阵位移法计算, 则与结点位移 1、2 (正方向见图虚线标示) 对应的等效结点荷载向量为 $\begin{pmatrix} -\frac{9q_1^2}{16} \\ \frac{q_1^2}{16} \end{pmatrix}$ 。(浙江大学 2012)

8-6 图 8-45 所示结构, 已知两杆 $EI = \text{常数}$ 。若按矩阵位移法求得结点位移 1、2 分别为 θ_1 、 Δ_2 , 则杆件 BC 左、右端的弯矩各为 $\frac{4EI}{L}\theta_1 - \frac{6EI}{L^2}\Delta_2 - \frac{qL^2}{12}$ 、 $\frac{2EI}{L}\theta_1 + \frac{6EI}{L^2}\Delta_2 - \frac{qL^2}{12}$ (均以下侧受拉为正)。(浙江大学 2012)

8-7 试求图 8-46 所示结构中结点 3 的综合结点荷载列阵 $F_3 = (X_3, Y_3, \theta_3)^T$ 。(中南大学 2009)



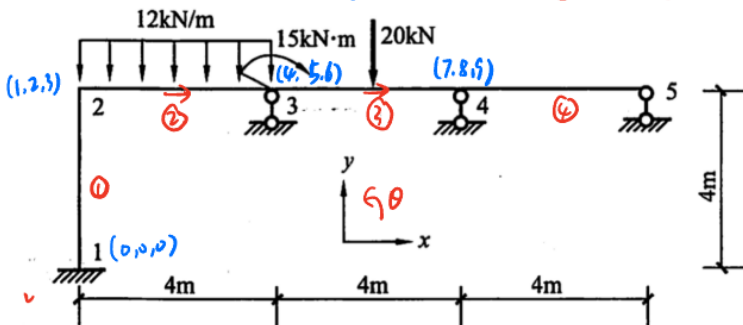
$$k \odot \odot + F_P \odot$$

$$= \begin{bmatrix} \frac{12EI}{l^3} & \frac{6EI}{l^2} & -\frac{12EI}{l^3} & -\frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{4EI}{l} & -\frac{6EI}{l^2} & -\frac{2EI}{l} \\ -\frac{12EI}{l^3} & -\frac{6EI}{l^2} & \frac{12EI}{l^3} & \frac{6EI}{l^2} \\ \frac{6EI}{l^2} & \frac{2EI}{l} & -\frac{6EI}{l^2} & \frac{4EI}{l} \end{bmatrix} \begin{bmatrix} 0 \\ \theta_1 \\ \theta_2 \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{ql^2}{2} \\ -\frac{ql^2}{12} \\ -\frac{ql^2}{2} \\ \frac{ql^2}{12} \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$

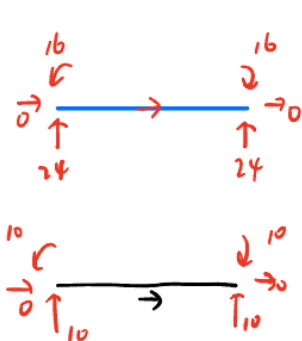
$$\begin{bmatrix} \frac{4EI}{l} \theta_1 - \frac{6EI}{l^2} \theta_2 - \frac{ql^2}{12} \\ \frac{2EI}{l} \theta_1 - \frac{6EI}{l^2} \theta_2 + \frac{ql^2}{12} \end{bmatrix}$$

$$\frac{4EI}{l} \theta_1 - \frac{6EI}{l^2} \theta_2 - \frac{ql^2}{12} = \frac{2EI}{l} \theta_1 - \frac{6EI}{l^2} \theta_2 + \frac{ql^2}{12}$$

注意：在矩阵位移法中，如果题目中完全没有刚度相关的数据，这种类型的题目，是需要考虑轴向变形的。



$$P^3 = P_E^3 + P_D^3$$



$$P_E^{\odot} = - \begin{bmatrix} 0 \\ 24 \\ 16 \\ 0 \\ 24 \\ -16 \end{bmatrix} = \begin{bmatrix} 0 \\ -24 \\ -16 \\ 0 \\ -24 \\ 16 \end{bmatrix}$$

$$P_E^{\odot} = - \begin{bmatrix} 0 \\ 10 \\ 10 \\ 10 \\ 0 \\ -10 \end{bmatrix} = \begin{bmatrix} 0 \\ -10 \\ -10 \\ -10 \\ 0 \\ 10 \end{bmatrix}$$



$$p_E = \begin{bmatrix} 0 \\ -34 \\ 6 \end{bmatrix} \quad p_D = \begin{bmatrix} 0 \\ 0 \\ -15 \end{bmatrix} \quad p_3 = \begin{bmatrix} 0 \\ -34 \\ -9 \end{bmatrix}$$

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8-8 按先处理法求图 8-47 所示连续梁的结点荷载列阵 P 。(西南交通大学 2007)

8-9 图 8-48 (a) 所示结构, 不考虑轴向变形, 整体坐标见图 8-48 (b), 图中括号内数码为结点总码 (力和位移按水平、竖直、转动方向顺序排列), 求等效结点荷载列阵 P 。(山东建筑大学 2008)

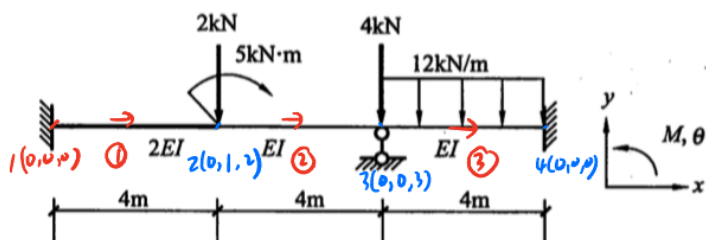


图 8-47

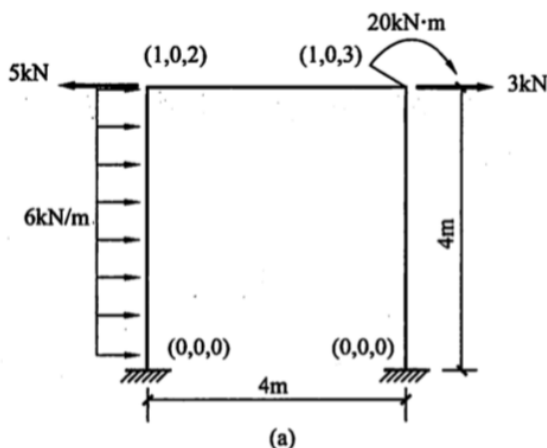
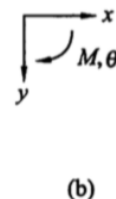


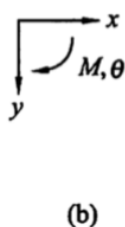
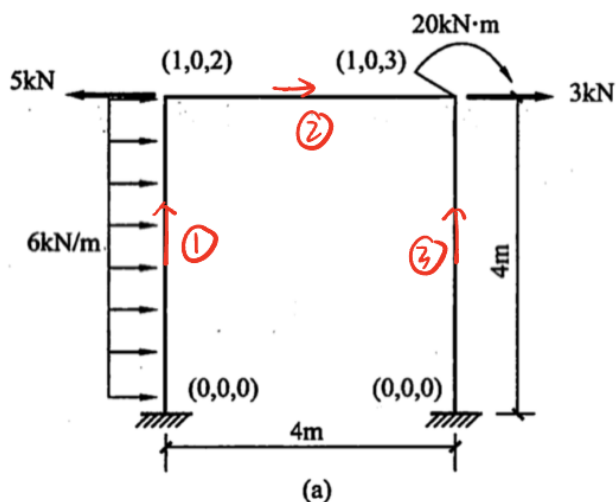
图 8-48



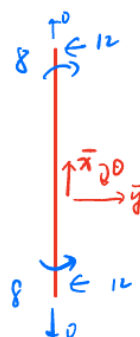
$$P_E^{(1)} = - \begin{bmatrix} 0 \\ 24 \\ 16 \\ 0 \\ 24 \\ -16 \end{bmatrix} = \begin{bmatrix} 0 \\ -24 \\ -16 \\ 0 \\ -24 \\ 16 \end{bmatrix}$$

$$P = \begin{bmatrix} -2 \text{ kN} \\ -5 \text{ kN} \\ -16 \text{ kN} \end{bmatrix}$$

$$P_E = \begin{bmatrix} 0 \\ 0 \\ -16 \end{bmatrix} \quad P_D = \begin{bmatrix} -2 \\ -5 \\ 0 \end{bmatrix}$$



注意: 等效节点荷载向量, 就是杆端力向量反号。



$$\alpha = -90^\circ, \cos \alpha = 0, \sin \alpha = -1$$



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$$\bar{P}_E^0 = - \begin{bmatrix} 0 \\ -12 \\ -8 \\ 0 \\ -12 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 12 \\ 8 \\ 0 \\ 12 \\ -8 \end{bmatrix}$$

$$P_E^0 = T^T \bar{P}_E^0 =$$

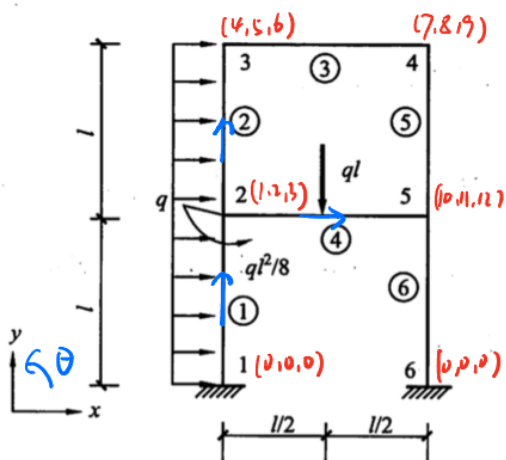
$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 12 \\ 8 \\ 0 \\ 12 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 8 \\ 12 \\ 0 \\ -8 \end{bmatrix}$$

$$P_E = \begin{bmatrix} 12 \\ -8 \\ 0 \end{bmatrix}, \quad P_D = \begin{bmatrix} -2 \\ 0 \\ 20 \end{bmatrix}, \quad P = \begin{bmatrix} 10 \text{ kN} \\ -8 \text{ kN}\cdot\text{m} \\ 20 \text{ kN}\cdot\text{m} \end{bmatrix}$$

$$18 - 6 = 12$$

8-10 图 8-49 所示平面结构用矩阵位移法计算，引入支承条件后的整体刚度矩阵为多少阶？求结点 2 和结点 5 的综合结点荷载列阵。（中南大学 2011）

4. 内力和位移的计算



8-11 已知图 8-50 所示连续梁的结点位移列阵 $\Delta = (-0.003 \quad 0.004 \quad 0.005)^T$ (分别为结点 2、3、4 的转角)，设各杆的 $E = 2 \times 10^4 \text{ kN/cm}^2$, $I = 400 \text{ cm}^4$, $l = 400 \text{ cm}$ 。求单元 ② 的剪力和弯矩。（华南理工大学 2010）

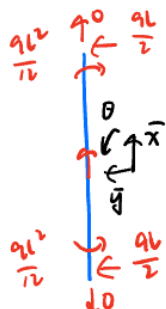
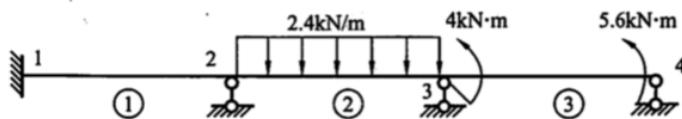


图 8-49

$$\bar{P}_e^0 = - \begin{bmatrix} 0 \\ \frac{ql}{2} \\ \frac{ql^2}{2} \\ 0 \\ \frac{ql}{2} \\ -\frac{ql^2}{2} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{ql}{2} \\ -\frac{ql^2}{2} \\ 0 \\ \frac{ql}{2} \\ \frac{ql^2}{2} \end{bmatrix}$$

$$P_e^0 = T^T \bar{P}_e^0 =$$

$$\begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{ql}{2} \\ \frac{ql^2}{2} \\ 0 \\ \frac{ql}{2} \\ -\frac{ql^2}{2} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{ql}{2} \\ -\frac{ql^2}{2} \\ 0 \\ \frac{ql}{2} \\ \frac{ql^2}{2} \end{bmatrix}$$

$$\alpha = 90^\circ, \quad \cos \alpha = 0, \quad \sin \alpha = 1$$

$$P_e^{\text{②}} = \begin{bmatrix} \frac{qL}{2} \\ 0 \\ \frac{qL^2}{12} \\ \frac{qL}{2} \\ 0 \\ \frac{qL^2}{12} \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix}$$

$$P_e^{\text{②}} = - \begin{bmatrix} 0 \\ \frac{qL}{2} \\ \frac{qL^2}{8} \\ 0 \\ \frac{qL}{2} \\ -\frac{qL^2}{8} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{qL}{2} \\ -\frac{qL^2}{8} \\ 0 \\ \frac{qL}{2} \\ \frac{qL^2}{8} \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 10 \\ 11 \\ 12 \end{matrix}$$

$$P_2^E = \begin{bmatrix} \frac{qL}{2} \\ -\frac{qL}{2} \\ -\frac{qL^2}{8} \\ 0 \end{bmatrix} \quad P_2^D = \begin{bmatrix} 0 \\ \frac{qL^2}{8} \end{bmatrix}$$

$$P_2 = \begin{bmatrix} \frac{qL}{2} \\ -\frac{qL}{2} \\ 0 \end{bmatrix} \quad P_3 = \begin{bmatrix} 0 \\ \frac{qL}{2} \\ \frac{qL^2}{8} \end{bmatrix}$$

$$\bar{F}^{\text{②}} \delta^{\text{②}} + \bar{F}_P^{\text{②}}$$

$$EI = 2 \times 10^4 \text{ kN/cm}^2 \times 400 \text{ cm}^4 \times 10^{-4}$$

$$EI = 800 \text{ kN} \cdot \text{m}^2$$

$$l = 4 \text{ m}$$

$$\frac{12EI}{l^3} = \frac{12 \times 800}{4^3} = 150$$

$$\frac{6EI}{l^2} = \frac{6 \times 800}{4^2} = 300$$

$$\frac{4EI}{l} = \frac{4 \times 800}{4} = 800$$

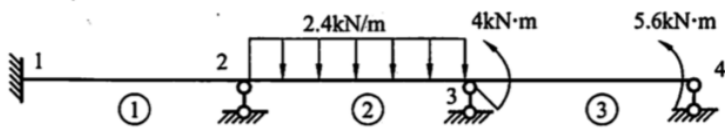
$$\begin{matrix} 3.2 & & 3.2 \\ \uparrow & & \uparrow \\ 4.8 & & 4.8 \end{matrix}$$

$$Q_{23} = 5.1 \text{ kN}$$

$$Q_{32} = -4.5 \text{ kN}$$

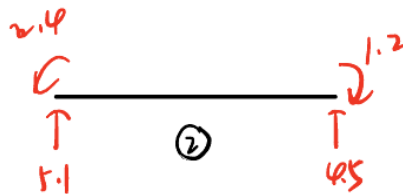
8-11 已知图 8-50 所示连续梁的结点

位移列阵 $\Delta = (-0.003 \quad 0.004 \quad 0.005)^T$ (分别为结点 2、3、4 的转角), 设各杆的 $E = 2 \times 10^4 \text{ kN/cm}^2$, $I = 400 \text{ cm}^4$, $l = 400 \text{ cm}$ 。求单元②的剪力和弯矩。(华南理工大学 2010)



$$\begin{bmatrix} 150 & 300 & -150 & 300 \\ 300 & 800 & -300 & 400 \\ -150 & -300 & 150 & -300 \\ 300 & 400 & -300 & 800 \end{bmatrix} \begin{bmatrix} 0 \\ -0.003 \\ 0 \\ 0.004 \end{bmatrix} + \begin{bmatrix} 4.8 \\ 3.2 \\ 4.8 \\ -3.2 \end{bmatrix}$$

$$= \begin{bmatrix} 5.1 \text{ kN} \\ 2.4 \text{ kN} \cdot \text{m} \\ 4.5 \text{ kN} \\ -1.2 \text{ kN} \cdot \text{m} \end{bmatrix}$$



8-12 采用矩阵位移法求图 8-51 所示桁架各杆内力。已知：EA=60kN， $F_P=100\text{kN}$ (上海大学 2008、武汉大学 2008 类似)

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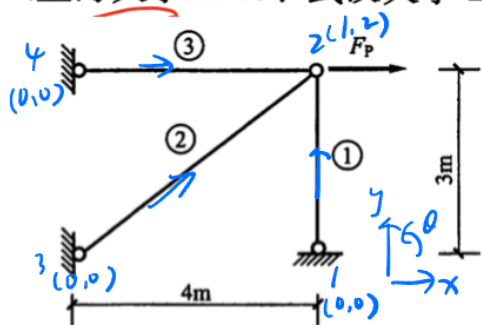


图 8-51

① $\alpha = 90^\circ, \cos\alpha = 0, \sin\alpha = 1$

$$k^{\textcircled{1}} = T^T \bar{k}^{\textcircled{1}} T = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 20 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 \\ -20 & 0 & 20 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 20 & 0 & -20 \\ 0 & 0 & 0 & 0 \\ 0 & -20 & 0 & 20 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 1 \\ 2 \end{matrix}$$

② $37^\circ, \cos\alpha = \frac{4}{5}, \sin\alpha = \frac{3}{5}$

$$k^{\textcircled{2}} = T^T \bar{k}^{\textcircled{2}} T = 12 \begin{bmatrix} \frac{16}{25} & \frac{12}{25} & -\frac{16}{25} & -\frac{12}{25} \\ \frac{12}{25} & \frac{9}{25} & -\frac{12}{25} & -\frac{9}{25} \\ -\frac{16}{25} & -\frac{12}{25} & \frac{16}{25} & \frac{12}{25} \\ -\frac{12}{25} & -\frac{9}{25} & \frac{12}{25} & \frac{9}{25} \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 1 \\ 2 \end{matrix}$$

$$k^{\textcircled{2}} = \begin{bmatrix} 15 & 0 & -15 & 0 \\ 0 & 0 & 0 & 0 \\ -15 & 0 & 15 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ 1 \\ 2 \end{matrix}$$



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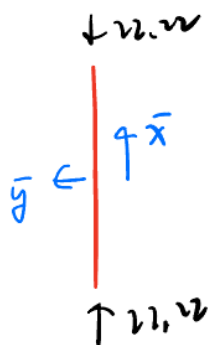
$$k = \begin{bmatrix} 22.68 & 5.76 \\ 5.76 & 24.32 \end{bmatrix}$$

$$\begin{bmatrix} 22.68 & 5.76 \\ 5.76 & 24.32 \end{bmatrix}$$

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$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix} = \begin{bmatrix} 4.6913 \\ -1.1111 \end{bmatrix}$$

$$\bar{F}_0 \bar{\Delta}^0 = \bar{F}_0^T \Delta^0 = \begin{bmatrix} 20 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 \\ -20 & 0 & 20 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 4.6913 \\ -1.1111 \end{bmatrix} = \begin{bmatrix} 22.22 \\ 0 \\ -22.22 \\ 0 \end{bmatrix}$$



$$\bar{F}_0 \bar{\Delta}^0 = \bar{F}_0^T \Delta^0 = \begin{bmatrix} 12 & 0 & -12 & 0 \\ 0 & 0 & 0 & 0 \\ -12 & 0 & 12 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{4}{5} & \frac{3}{5} & 0 & 0 \\ -\frac{3}{5} & \frac{4}{5} & 0 & 0 \\ 0 & 0 & \frac{4}{5} & \frac{3}{5} \\ 0 & 0 & -\frac{3}{5} & \frac{4}{5} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 4.6913 \\ -1.1111 \end{bmatrix}$$

$$= \begin{bmatrix} -37.04 \\ 0 \\ 37.04 \\ 0 \end{bmatrix}$$

Diagram showing a horizontal member with a vertical force F-bar at the left end. The member is labeled with a horizontal axis and a vertical axis. The force F-bar is shown as a blue arrow pointing upwards. The member is labeled with a horizontal axis and a vertical axis. The force F-bar is shown as a blue arrow pointing upwards.



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$$\bar{k} \bar{\delta} = \begin{bmatrix} 15 & 0 & -15 & 0 \\ 0 & 0 & 0 & 0 \\ -15 & 0 & 15 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 4.6913 \\ -1.1111 \end{bmatrix} = \begin{bmatrix} -70.27 \\ 0 \\ 70.27 \\ 0 \end{bmatrix}$$

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$$\begin{matrix} \leftarrow & \xrightarrow{\bar{x}} & \rightarrow \\ 70.27 & & 70.27 \end{matrix}$$

8-13 图 8-52 所示结构，已知结点 2 的水平、竖向和转角位移分别为 0、-0.04、-0.1667，结点 3 的转角为 0.2220，试求 23 杆 2 端杆端力。已知单元刚度矩阵中的部分



元素:

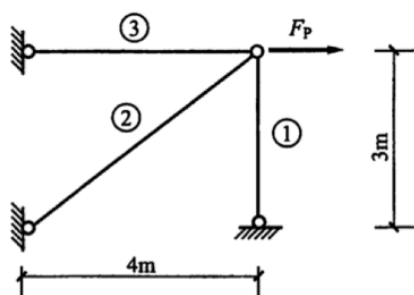


图 8-51

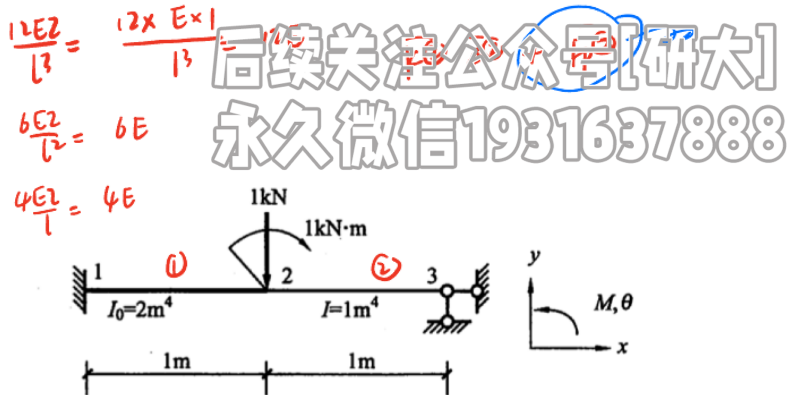


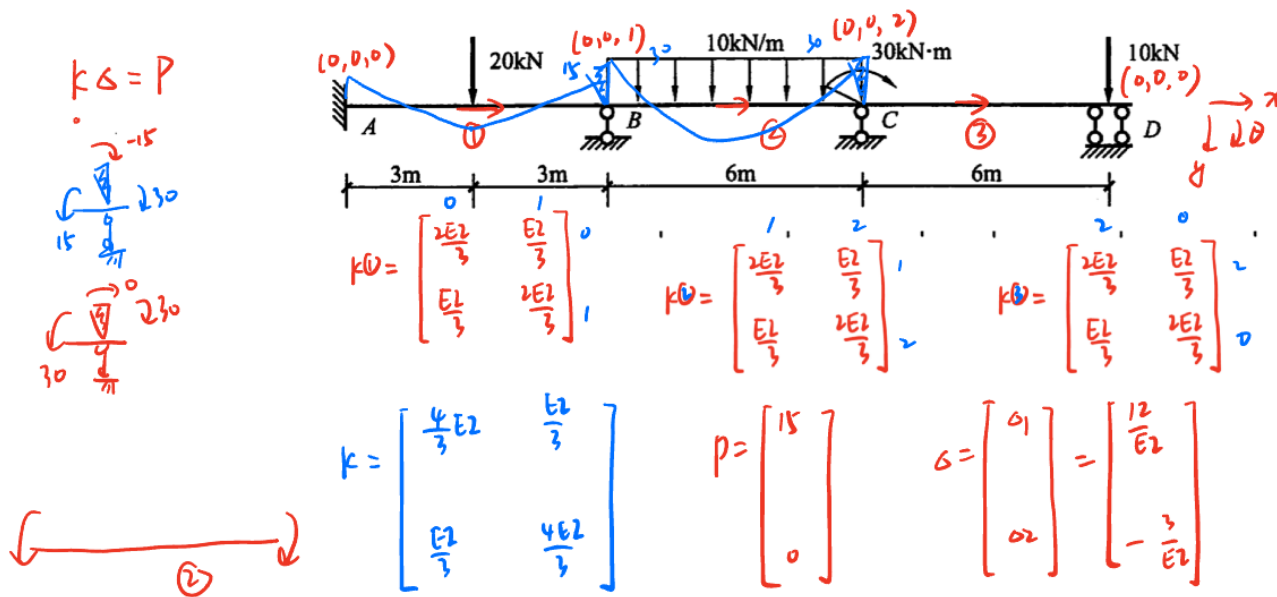
图 8-52

$$k_{22} = -k_{52} = k_{55} = \frac{12EI}{l^3}, \quad k_{32} = k_{62} = -k_{53} = -k_{65} = \frac{6EI}{l^2}, \quad k_{33} = 2k_{63} = k_{66} = \frac{4EI}{l}.$$

学 2008)

$$\begin{bmatrix} 12E & 6E & -12E & 6E \\ 6E & 4E & -6E & 2E \\ -12E & -6E & 12E & -6E \\ 6E & 2E & -6E & 4E \end{bmatrix} \begin{bmatrix} -0.0926 \\ -0.166 \\ 0 \\ 0.2220 \end{bmatrix} = \begin{bmatrix} -0.7794E \\ -0.7784E \\ 0.7794E \\ -0.001E \end{bmatrix}$$

8-14 用矩阵位移法计算并作图 8-53 所示连续梁的弯矩图。各杆 EI 相同，为常数。
(重庆大学 2009)



注意：如果只要求求解杆端弯矩，单元刚度矩阵可以写成2阶的。

$$\begin{bmatrix} \frac{2EI}{3} & \frac{EI}{3} \\ \frac{EI}{3} & \frac{4EI}{3} \end{bmatrix} \begin{bmatrix} \frac{12}{EI} \\ -\frac{3}{EI} \end{bmatrix} + \begin{bmatrix} -10 \\ 30 \end{bmatrix} = \begin{bmatrix} -23 \\ 32 \end{bmatrix}$$

